



The next big step

An outline of the plans to build a quantum internet where data cannot be intercepted.
<https://digit.in/jul20-01>



Quantum backbone

A quantum backbone link has been established between Beijing and Shanghai.
<https://digit.in/jul20-02>

The Unhackable Internet

An ambitious project to make internet data security impenetrable

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he internet we are so wildly acclimated to today is a virtual playground for opportunist hackers.

Issues such as insecure communication links and unsafe data in the cloud are major vulnerabilities that are regularly exploited, often leading

to catastrophic consequences. Although the internet has grown leaps and bounds from the time of its inception, with gargantuan amounts of data transferred daily all over the world, its weaknesses are vast as well. The answer to this seems to be an 'unhackable' internet or a quantum internet that is being developed by several quantum physicists and researchers around the world. This 'unhackable' internet is freed from the shackles of traditional networks and could provide the highest level of privacy and security. It could be used to send unhackable messages and data, improve the accuracy of GPS and even enable cloud-based quan-

tum computing in the future. MIT Technology Review named the 'unhackable' internet one of the top 10 technological breakthroughs in the year 2020. Let's study the quantum internet and the progress made so far in the technology.

QUBITS: ITS STRENGTHS, AND ITS LIMITATIONS

To figure out why the quantum internet is possibly the ultimate level of internet security, you will have to understand why the internet we have today is not. The internet, as we know it today, is quite vulnerable to hackers since data still travels along cables in the form of bits. Bits are the lowest quantification of data. These bits are a stream of electrical or optical pulses that travel along cables, represented in 1s and 0s. If a hacker does manage to tap into the cables, she/he will be able to read the data as well as copy it.

However, the laws of quantum physics, which apply to the 'unhackable' quantum internet as well, allow a bit (could be an atom, an electron, or a photon of light) to occupy something known as the quantum state. In the quantum state, the bit is represented by a combination of 1 and 0 simultaneously. This particle is called a quantum bit or a 'qubit' and the phenomenon of being in this rapidly-changing state is called 'superposition' in quantum physics. Why is this useful? Well, if a hacker does manage to tap into the cables hosting the stream of qubits, they 'collapse' out of superposition, into either 1 or 0. The hacker thereby destroys the quantum information that was tampered with and also leaves a clear indicator to alert people of the intrusion.

Due to this innate property of qubits, they have been used extensively for quite some time now to generate encryption keys. The process of generating encryption keys using qubits is known as quantum key distribution (QKD). In this method, data is sent over a classical form of network while the decryption keys are transmitted separately in a quantum